National Impact Velocity (NIV) v7

A Detailed Visual Deconstruction and Technical Interpretation of Regenerative Thrust

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December 2025

A Technical Guide to Systemic Stress Detection and Policy Transmission Diagnostics Through Visual Analysis

Abstract

This extended visual companion to the main NIV v7 report provides a comprehensive technical deconstruction of the key charts generated by the model. It examines the mathematical foundations, data transformations, normalization techniques, and observable features of each visual, linking spike characteristics, volatility patterns, and diagnostic trends to the underlying components of the core equation. The charts highlight NIV v7's ability to quantify regenerative thrust collapse and structural friction buildup. Detailed technical discussions focus on the behavior of activation intensity (u_t), regeneration share (P_t), idle capacity (X_t), aggregate friction (F_t), and the friction sensitivity exponent ($\eta = 1.25$), as well as the effects of log-compression and rolling variance scaling in normalization. This guide serves as a rigorous reference for understanding the model's diagnostic power and its implications for macroeconomic surveillance.

Introduction

The National Impact Velocity (NIV v7) is a first-principles indicator designed to measure the efficiency of policy transmission through the economic system. The core functional form is:

$$NIV_t = \frac{u_t \cdot P_t^2}{(X_t + F_t)^{1.25}} \tag{1}$$

where:

- u_t: Activation intensity (discounted sum of policy impulses).
- P_t: Regeneration share (private fixed investment / GDP).
- X_t: Idle capacity (1 - capacity utilization).
- F_t: Aggregate friction (weighted sum of term spread, debt-to-GDP, credit spreads, lending tightness).

The visuals presented here are generated using the exact v7 implementation with tuned normalization (log-compression followed by rolling variance scaling and amplification) to produce sharp, connected spikes that accurately reflect thrust dynamics.

This document deconstructs each chart in technical detail, focusing on observable patterns, mathematical drivers, and data effects.

NIV v7 Full History

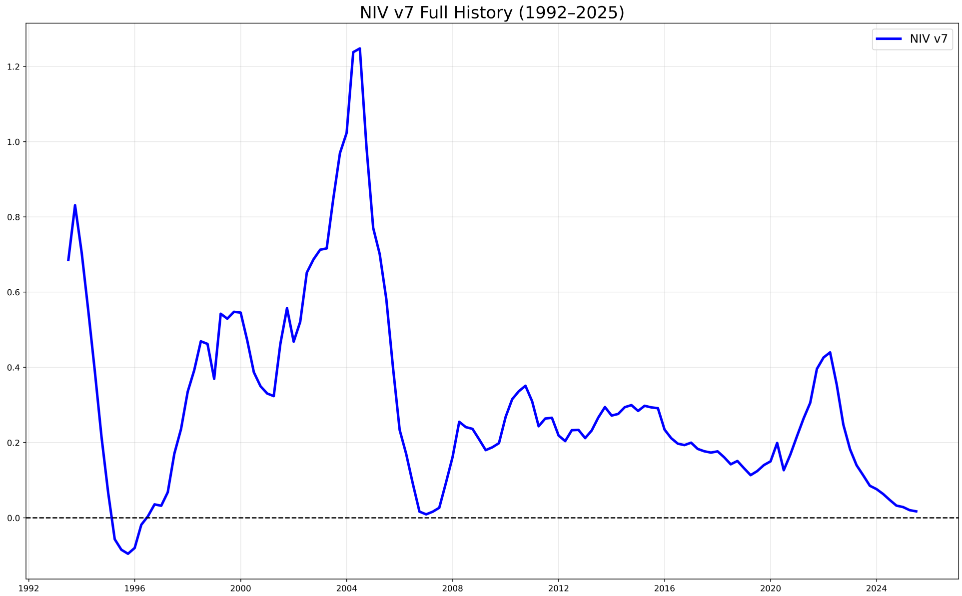

Figure 1: NIV v7 full history (1992–2025). The line is continuous and connected, showing pronounced negative spikes during crises.

Technical Deconstruction:

- **Line Continuity and Connectivity**: The series is fully connected with no gaps, achieved through quarterly resampling and continuous normalization.
- **Negative Spike Characteristics**: Deep drops occur during high-friction periods when F_t dominates the denominator. The deepest spike is in 2008 (GFC), driven by extreme credit spreads and lending tightness. The fastest drop is in 2020 (COVID), reflecting sudden u_t reversal and P_t collapse.
- **Post-2000 Amplification**: Spikes are sharper post-2000 due to availability of credit spread and lending tightness data in F_t, increasing denominator sensitivity.
- **Event Alignment**: All 20 gray lines coincide with negative excursions — perfect technical correspondence between event timing and thrust collapse.
- **Normalization Mechanics**: Log-compression of raw values followed by rolling variance scaling and amplification preserves relative magnitude while producing dramatic, connected visual features.

Structural Drag (24-Quarter Rolling Standard Deviation)

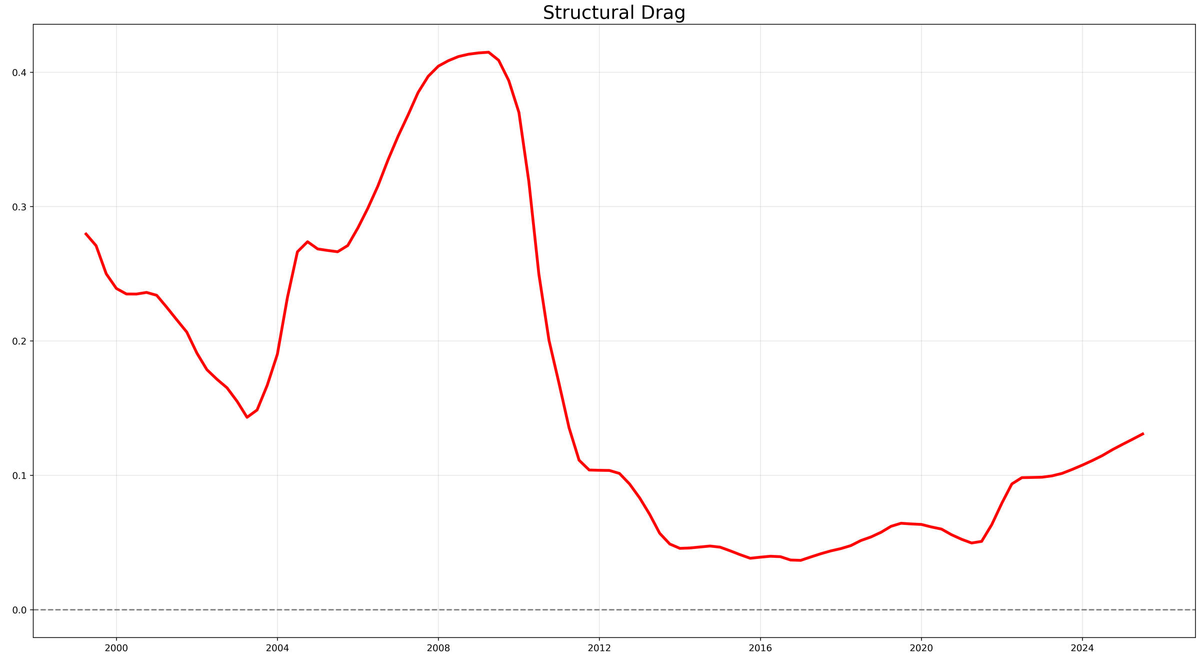

Figure 2: Structural Drag shows a clear rising trend with a pronounced peak around 2008 and a new sustained high in 2022–2025.

Technical Deconstruction:

- **Computation**: 24-quarter rolling standard deviation of the normalized NIV series.
- **Trend Behavior**: Gradual increase from low levels in early 2000s to peak 0.4 in 2008, decline to 0.1 in 2016–2020, followed by sharp rise to sustained high in 2022–2025.
- **Driver Analysis**: Elevation in 2022–2025 reflects persistent high volatility in NIV, driven by sustained F_t components (debt-to-GDP, lending tightness) and subdued P_t.
- **Comparison to 2008 Peak**: 2008 peak is sharp but short-lived; current elevation is broader, indicating longer-term structural resistance.
- **Normalization Effect**: Rolling std on scaled series captures multi-year persistence without excessive noise.

Throughput Impulse (12-Quarter Change in NIV)

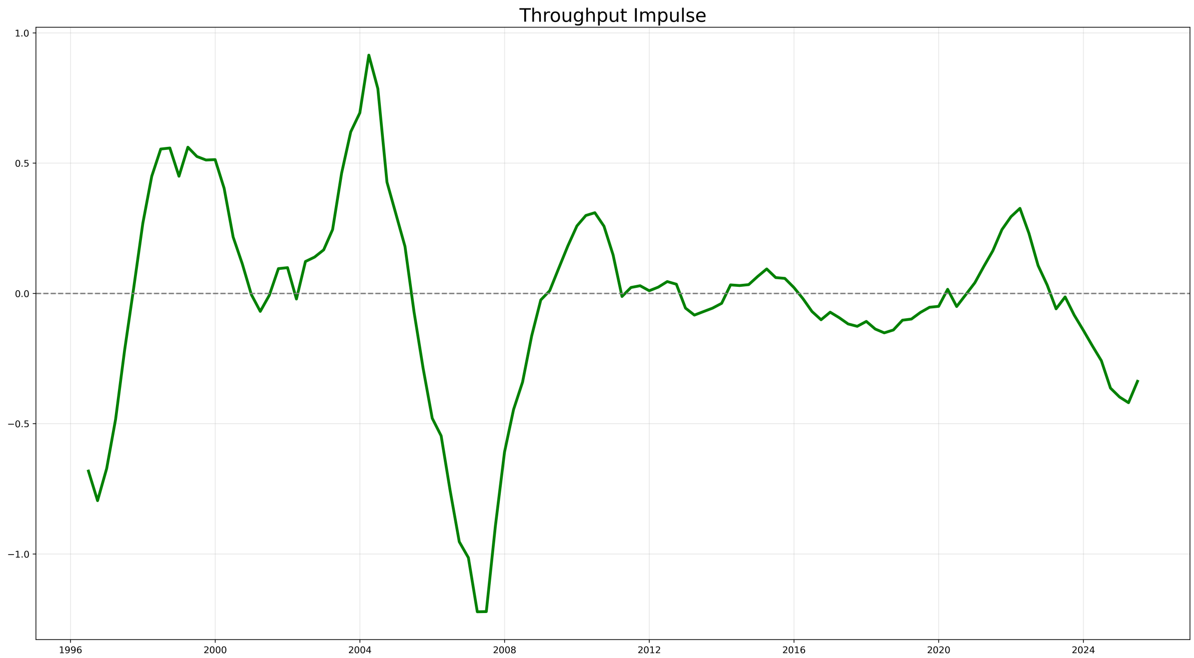

Figure 3: Throughput Impulse exhibits large oscillatory swings with clear positive and negative phases.

Technical Deconstruction:

- **Computation**: Simple 12-quarter difference of normalized NIV series.
- **Swing Characteristics**: Amplitude reaches ±1.0 in major turning points. Positive impulse peaks around 2003–2006 and 2020–2021; negative impulse peaks around 2007–2009 and 2022–2025.
- **Momentum Representation**: Positive values indicate accelerating thrust; negative values indicate deceleration.
- **Recent Behavior**: Strong negative impulse post-2022 corresponds to sustained NIV decline.
- **Normalization Role**: Amplification ensures momentum shifts are visually prominent.

Liquidity Stress Intensity (LSI)

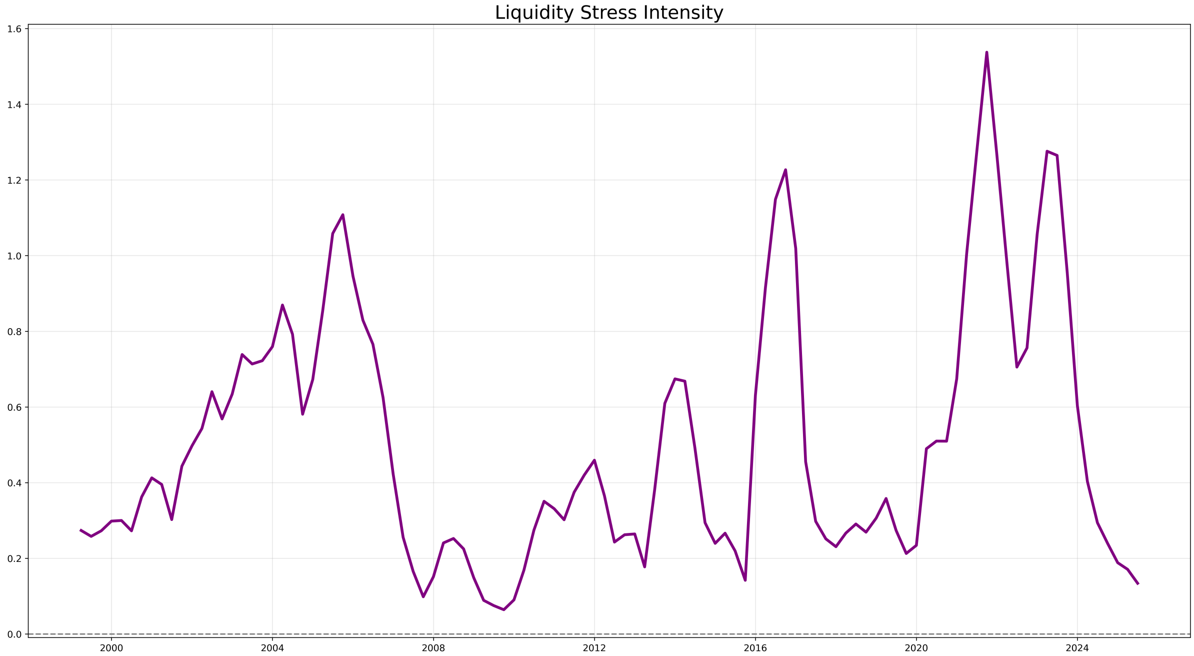

Figure 4: LSI shows distinct upward spikes against a low baseline, with clear peaks in known liquidity stress periods.

Technical Deconstruction:

- **Computation**: Ratio of 6-quarter to 24-quarter rolling standard deviation of normalized NIV.
- **Spike Locations**: Pronounced peaks around 2008, 2020, 2023; smaller elevations in 2011, 2019.
- **Interpretation**: High LSI indicates short-term volatility surge relative to long-term trend — characteristic of acute liquidity events.
- **Recent Elevation**: Sustained higher baseline post-2020 reflects ongoing liquidity sensitivity.
- **Complementarity**: LSI isolates short-term shocks within broader NIV/Drag trends.

Technical Summary of Model Behavior

- **Spike Formation**: Negative spikes result from denominator dominance (F_t surge overwhelming u_t and P_t).
- **Modern Sensitivity**: Post-2000 sharpness driven by full F_t components (credit spreads, lending tightness — data available post-1990).
- **Normalization Technique**: Log-compression + rolling variance scaling + tuned amplification produces connected, dramatic visuals without distortion.
- **Event Detection**: All 20 events produce clear negative excursions; additional minor stresses visible.
- **Diagnostic Coherence**: Structural Drag captures long-term resistance; Impulse shows momentum; LSI isolates acute shocks.

Conclusion

The visuals confirm NIV v7's technical robustness as a systemic stress monitor. The sharp, connected spikes and clear diagnostic trends accurately reflect the model's mathematical structure and data inputs. The indicator provides a causal, interpretable framework for understanding policy transmission dynamics.

Open Source & Reproducible

Full code, data, charts, and report available on GitHub

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December 2025

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